

Hands-on 1 – Creating a Spring Web Project using Maven

1. Project Execution and Output Verification

1.1 Maven Build Output

The project was packaged using the Maven build tool. Since the setup was completed on a personal network, the dependencies were fetched directly without routing through a corporate proxy firewall.

Terminal Build Command:

mvn clean package

Output:

```
PS D:\SD\Digital-Nurture-JavaFSE-main\local repo\Cognizant-exercices\Week 2\Spring Rest (Spring Boot 3)\spring-learn\spring-learn> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.cognizant:spring-learn >-----
[INFO] Building 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] Replacing main artifact D:\SD\Digital-Nurture-JavaFSE-main\local repo\Cognizant-exercices\Week 2\Spring Rest (Spring Boot 3)\spring-learn\spring-learn\target\spring-learn-0.0.1-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to D:\SD\Digital-Nurture-JavaFSE-main\local repo\Cognizant-exercices\Week 2\Spring Rest (Spring Boot 3)\spring-learn\spring-learn\target\spring-learn-0.0.1-SNAPSHOT.jar.original
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 18.935 s
[INFO] Finished at: 2026-06-29T18:02:32+05:30
[INFO] -----
```

2. The Spring Boot Console Execution Logs

When you open SpringLearnApplication.java and click Run, your console will spin up the Spring Boot framework. This is where your custom logging statements prove the main() method executed correctly. What your console output should look like:

```
PS D:\SD\Digital-Nurture-JavaFSE-main\local repo> & 'C:\Program Files\Eclipse Adoptium\jdk-25.0.3.9-hotspot\bin\java.exe' '@C:\Users\adity\AppData\Local\Temp\c_dynm4b5rn8kj5auqvxxzx019df.argfile' 'com.cognizant.spring_learn.SpringLearnApplication'
18:06:36.261 [main] INFO com.cognizant.spring_learn.SpringLearnApplication -- --- STARTING: Entering main() method of SpringLearnApplication ---
18:06:36.399 [restartedMain] INFO com.cognizant.spring_learn.SpringLearnApplication -- --- STARTING: Entering main() method of SpringLearnApplication ---

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:: Spring Boot ::                (v4.1.0)

2026-06-29T18:06:36.660+05:30 INFO 2576 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 25.0.3 with PID 2576 (D:\SD\Digital-Nurture-JavaFSE-main\local repo\Cognizant-exercises\Week 2\Spring Rest (Spring Boot 3)\spring-learn(spring-learn\target\classes started by adity in D:\SD\Digital-Nurture-JavaFSE-main\local repo)
2026-06-29T18:06:36.662+05:30 INFO 2576 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-06-29T18:06:36.694+05:30 INFO 2576 --- [spring-learn] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set et 'spring.devtools.add-properties' to 'false' to disable
2026-06-29T18:06:36.694+05:30 INFO 2576 --- [spring-learn] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-06-29T18:06:37.171+05:30 INFO 2576 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-06-29T18:06:37.180+05:30 INFO 2576 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-06-29T18:06:37.181+05:30 INFO 2576 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-06-29T18:06:37.205+05:30 INFO 2576 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 510 ms
2026-06-29T18:06:37.405+05:30 INFO 2576 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-06-29T18:06:37.409+05:30 INFO 2576 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 1.007 seconds (process running for 1.263)
2026-06-29T18:06:37.411+05:30 INFO 2576 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : --- SUCCESS: SpringBoot Container is fully initialized and running ---
```